Jacob Pilawa

Astrophysics Ph.D. Candidate, UC Berkeley

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https://jacobpilawa.github.io/

Education

- **Ph.D., Astrophysics, University of California, Berkeley** — Berkeley, CA
  May 2026, Advisor: Chung-Pei Ma
- **M.A., Astrophysics, University of California, Berkeley** — Berkeley, CA
  August 2020 – May 2022
- **B.A., Astrophysics (with honors), Colgate University** — Hamilton, NY
  August 2016 – May 2020, *summa cum laude*

Specialized Skills and Software

Programming Languages: Python (Advanced), SQL (Intermediate), R (Intermediate), MATLAB (Intermediate), Mathematica (Intermediate)

Computing/Software: `numpy`, `scipy`, `scikit-learn`, `pandas`, `streamlit`, `matplotlib`, Linux, bash, LaTeX, git, SLURM, Microsoft Office Suite

Work Experience

Graduate Student Researcher, UC Berkeley — **2020 – Current**

- Lead the discovery and characterization of 3 new supermassive black holes out of the ∼100 that currently exist. Involved measuring stellar kinematics from high-resolution infrared spectroscopy and imaging, fitting galactic light-profiles using 2D image processing software and fitting routines, and statistically modeling large amounts ($\geq$ 1 TB) of data to extract supermassive black hole masses via machine learning and Monte Carlo methods.
- Developed and implemented a Bayesian statistical framework to test the accuracy and precision of results coming from large "black-box" galaxy simulations, ensuring that the black holes previously discovered are robust. Involved creating a large number of "mock" galaxy models with known parameters using regularized fitting routines, with which we can test for accurate parameter recovery from our code.

Modeling and Simulations Engineer, The Aerospace Corporation — **May 2019, 2020 – August 2019, 2020**

- On-site (2019) and remote (2020) internship positions in the Photonics and Microtechnologies Departments at The Aerospace Corporation in El Segundo, California working on reservoir computing and photonic neural networks.
- Wrote MATLAB software which interfaced hardware (infrared lasers and spatial light modulators) with software (echo state network calculations) to perform classification of time-domain waveforms using speckle-based optical computing, with applications for in-situ signal processing.
- Performed theoretical tests relating the singular value spectrum and sparsity of echo state networks to their performance, in terms of memory capacity and robustness of predictions.

Undergraduate Research Assistant, Colgate University — **August 2019 – May 2020**

- Helped develop and test a mathematical framework for the gravitational capture of dark matter particles by stars and its effects on stellar evolution with advisor Cosmin Ilie, culminating in my undergraduate thesis work.
- Involved both pen-and-paper theoretical physics work, as well as writing custom MATLAB and Python code to compute dark matter capture rates by stars for a variety of realistic scenarios which can be used to test our theories against real observations

REU Summer Research Fellowship, University of Wyoming — **May 2018 – August 2018**

- National Science Foundation funded research position at the University of Wyoming under Daniel Dale.
- Planned observations for and operated the 2.3-meter Wyoming Infrared Observatory telescope over 30+ nights, collecting multi-wavelength observations of nearby galaxies to study their assembly and star formation histories via their spectral energy distributions
- Developed both an IRAF and Python pipeline for the image processing, data extraction, and statistical analysis of the galaxy photometric data. Presented the work at an international astrophysics conference in January 2019.

USA Jigsaw Puzzle Association Statistician — **Year-Round Volunteer**

- Leading the development and testing of the USA Jigsaw Puzzle Association Rating (JPAR), a numerical ranking system for competitive jigsaw speed puzzlers, along with the data collection, cleaning, and maintenance.
- A (soon-to-be migrated) prototype of this system called `Puzzleboard` is currently live and in use by ∼250 monthly users.

Publications

10. **J. D. Pilawa**, E. R. Liepold, C.-P. Ma, "A Ten-Billion-Solar-Mass Supermassive Black Hole in the Isolated Elliptical Galaxy NGC 57," in prep., to be submitted October 2025.
9. **J. D. Pilawa**, E. R. Liepold, C.-P. Ma, J. L. Walsh, J. E. Greene, "The MASSIVE Survey. XX. A Triaxial Stellar Dynamical Measurement of the Supermassive Black Hole Mass and Intrinsic Galaxy Shape of Giant Radio Galaxy NGC 315," The Astrophysical Journal, Published 2025 Aug 7. [link]
8. **J. D. Pilawa**, E. R. Liepold, C.-P. Ma, "TriOS Schwarzschild Orbit Modeling: Robustness of Parameter Inference for Masses and Shapes of Triaxial Galaxies with Supermassive Black Holes," The Astrophysical Journal, Published 2024 May 9. [link]
7. **J. D. Pilawa**, E. R. Liepold, S. C. Delgado-Andrade, J. L. Walsh, C.-P. Ma, M. E. Quenneville, J. E. Greene, J. P. Blakeslee, "The MASSIVE Survey - XVII. A Triaxial Orbit-based Determination of the Black Hole Mass and Intrinsic Shape of Elliptical Galaxy NGC 2693," The Astrophysical Journal, Published 2022 April 7. [link]
6. M. Ashner, U. Paudel, M. Luengo-Kovac, **J. Pilawa**, T. Shaw, and G. Valley, "Photonic reservoir computer using speckle in multimode waveguide ring resonators," Opt. Express 29, 19262-19277 (2021). [link]
5. C. Ilie, C. Levy, **J. Pilawa**, S. Zhang, "Probing below the neutrino floor with the first generation of stars." Submitted to Physical Review Letters, (2021). [link]
4. C. Ilie, C. Levy, **J. Pilawa**, S. Zhang, "Constraining Dark Matter properties with the first generation of stars." Submitted to Physical Review D, (2021). [link]
3. C. Ilie, **J. Pilawa**, S. Zhang, "Comment on 'Multiscatter stellar capture of dark matter."' Physical Review D, Volume 102, Issue 4, article id.048301 (2020). [link]
2. D. Dale, K. Anderson, L. Bran, I. Cox, C. Drake, N. Lee, **J. Pilawa**, F. Slane, S. Soto, ... ,"Radial Star Formation Histories in 32 Nearby Galaxies." The Astronomical Journal 159.5 (2020): 195. [link]
1. U. Paudel, M. Luengo-Kovac, **J. Pilawa**, T. Shaw, and G. Valley, "Classification of time-domain waveforms using a speckle-based optical reservoir computer," Opt. Express 28, 1225-1237 (2020). [link]

Teaching Experience

Graduate Student Instructor, Astro C202, *Astrophyiscal Fluid Dynamics*, UC Berkeley	2023, 2024, 2025
Graduate Student Instructor, Astro C228, *Extragalactic Astrophysics and Cosmology*, UC Berkeley	2025
Graduate Student Instructor, Astro C12, *The Planets*, UC Berkeley	2021, 2022, 2023
Graduate Student Instructor, Astro 120, *Optical and Infrared Lab*, UC Berkeley	2021, 2022
Graduate Student Instructor, Astro C10, *Introduction to General Astronomy*, UC Berkeley	2020, 2025
Co-Instructor, University Studies 350, *Design Lab*, Colgate University	2019, 2020

Talks & Posters

10. **J. Pilawa**, "Stellar Dynamical Mass Measurements and Intrinsic Shapes of MASSIVE Elliptical Galaxies," Texas A&M University, Department of Physics & Astronomy Extragalactic Group. [link to talk] February 2023
9. **J. Pilawa**, "Stellar Dynamical Mass Measurements of Massive Elliptical Galaxies," UC Berkeley Department of Astronomy Lunch Talk. [link to talk] March 2022
8. **J. Pilawa**, U. Paudel, M. Luengo-Kovac, G. Valley, J. Shaw, H. Doyle, M. Ashner, "Applications and Performance of Echo State Networks," Photonics Department, The Aerospace Corporation. [link to talk] August 2020
7. **J. Pilawa**, "Astrophysical Sources as Dark Matter Detectors," Honors Undergraduate Thesis Defense, Colgate University. [link to talk], [link to thesis] May 2020
6. U. Paudel, M. Luengo-Kovac, **J. Pilawa**, G. C. Valley, T. J. Shaw, "Reservoir computer using speckle in a multimode waveguide," Invited Talk, Photonics West, San Francisco, CA. February 2020
5. **J. Pilawa**, "Multi-scatter capture of intermediate mass dark matter in Pop. III stars," Undergraduate Thesis Defense, Colgate University. [link to talk] December 2019
4. **J. Pilawa**, U. Paudel, M. Luengo-Kovac, G. Valley, J. Shaw, "Speckle-based Reservoir Computing and Echo State Networks," Photonics Department, The Aerospace Corporation. [link to talk] August 2019
3. **J. Pilawa**, et al., "EDGES: Radial Star Formation Histories of NGC4143 and UGC07639,"Poster Presentation (Galaxy Evolution), AAS 223, Seattle, Washington. [link to poster] January 2019
2. **J. Pilawa**, et al., "EDGES: Radial Star Formation Histories of NGC4143 and UGC07639," Poster Presentation, KNAC Fall Symposium, Middlebury, VT. [link to poster] September 2018
1. **J. Pilawa**, K. Eckart, R. Stahlin, "The 2015-2016 Optical Outburst and Historic Light Curve of Blazar OJ287," KNAC Fall Symposium, Colgate University, New York. [link to poster] September 2017